

MANAGEMENT INTERVENTIONS FOR 119 PRIORITY ANIMAL SPECIES

This table identified priority management interventions for 119 animal species identified as high priority for urgent management intervention following the 2019-20 bushfires. Developed by the Wildlife and Threatened Species Bushfire Recovery Expert Panel, this analysis of actions will help guide post-fire management and conservation responses. Actions were based on consensus opinion across many experts and are intended to inform the management response immediately, and in the first year, post bushfire. While these actions are intended to support the management response for bushfire-affected species, this information should not be the only consideration of decision makers when focusing investment and effort.

^a The likely priority actions have been based on the analysis of species' traits. Some activities will require permits and permission of the public land manager.

^b Survey to establish extent of population loss, and establish baseline for ongoing monitoring.

^c This includes careful management of unburnt areas within or adjacent to recently burnt ground that provide refuges; also unburnt areas that are not adjacent to burnt areas. Includes protection from extensive, intense fire.

^d These columns indicate whether the actions are potentially suitable and feasible in the first year after fire, and record what action have taken place to date. In most cases, opportunities to salvage individuals and populations as a 'rescue' action are now past, but there may still be opportunities to capture wild animals to stock captive breeding programs, to carry out wild-to-wild translocations to unburnt habitat and to safe havens (fenced areas and islands), in future years and these are noted for some species.

^e Careful risk assessment is required in all cases.

^f Where appropriate, note that food and water provision may be less relevant as time goes on.

^g Note that many forest-dependent fauna are disadvantaged by post-fire salvage logging, especially hollow-dependent species, so an action to limit post-fire salvage logging will benefit most species. As well as priority actions for the first year post-fire, some key longer-term actions are indicated for some species.

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SPRAT ID	Species identified as the highest priority for urgent management action ^a		Rapid on-ground assessment for species and communities of concern ^b	Careful management of unburnt areas ^c	Emergency salvage of plant and animal species for ex-situ conservation or wild-to-wild translocation ^d		Supplementary shelter, food, and water for animals ^f	Feral predator and herbivore control to reduce the pressure on native species where appropriate		Longer-term additional actions ^g
	Common name	High scores for ecological traits			Salvage/capture into ex situ ^e	Salvage/translocate to nearby unburnt habitat or havens (including waterways for fish) ^e	Provision of supplementary resources	Exclude herbivores from unburnt and regenerating vegetation	Control introduced predators	
Invertebrates										
86868	<i>Pseudococcus markharveyi</i> Banksia montana mealybug	Herbivore with a specialised diet (one species of Banksia); habitat specialist that needs long-unburnt habitat; social	Yes	Yes	Unlikely	No, 100% of adult host plant population impacted	Unlikely	Yes	Unknown	
89125	<i>Bertmainius colonus</i> Eastern Stirling Range Pygmy Trapdoor Spider	High fire mortality because burrows are shallow	Yes	Yes	Yes	Maybe	Unlikely	Yes; to prevent trampling	Unknown	
87805	<i>Trioza barrettae</i> Banksia brownii plant louse	Herbivore with a specialised diet (one species of Banksia)	Yes	Yes	Yes	Maybe, if unburnt host plants exist, which does not already support louse	Unlikely	Yes	Unknown	
25289	<i>Thaumatoperla alpina</i> Alpine Stonefly		Yes	Yes	Unlikely	Unlikely	No	Yes; exclude from waterways	Unknown	

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26335	<i>Paralucia spinifera</i> Bathurst Copper Butterfly	Specialist herbivore with symbiotic relationship to a plant and an ant	Yes	Yes	No	Maybe, cool burns have been trialled in NSW which has established new habitat in some areas, possible target for nearby salvage	Maybe; possible habitat supplementation for ants	Yes	Yes; pigs, especially in unburnt habitat	- Weeds also a threat - Manage symbionts strong symbiotic relationships with <i>Bursaria spinosa</i> subsp <i>lasiophylla</i> (for egg laying) and with <i>Anonychomyrma itinerans</i> (ant) which protects larvae and receives some food form them - Somewhat dependent on fire, as species prefer young <i>Bursaria</i> plants
Mammals										
87634	<i>Sminthopsis griseoventer aitkeni</i> Kangaroo Island Dunnart	High mortality in intense fires; very high susceptibility to introduced predators (compounded by cover loss)	Yes	Yes	Yes, noting extreme difficulty of capturing live individuals in this rare species that is not attracted to baits, and that only very small numbers are left, probably confined to unburnt fragments	Maybe (species is so naturally rare, that unburnt patches may be unoccupied)	Yes; artificial shelters in burnt areas are an option	Yes	Yes; the most important action in the short to medium term.	
98	<i>Pseudomys oralis</i> Hastings River Mouse, Koontoo	High mortality in intense fires; very high susceptibility to introduced predators (compounded by cover loss)	Yes	Yes	Maybe; if founders required for later reintroductions	Maybe; as it is known to be absent from patches that seem suitable	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	- Mass regeneration of eucalypts and wattle post fire could suppress the grass they need, so localised control of these plants may be an option
217	<i>Potorous longipes</i> Long-footed Potoroo	Specialised habitat; very high susceptibility to introduced predators (compounded by cover loss)	Yes	Yes	Maybe; if founders required for later reintroductions	Maybe; nearby unburnt habitat may be saturated, but known to be absent from some sites in NSW that previously supported this species, so these locations may be options	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	- Cross jurisdiction workshop and analysis to see if wild to wild translocations to other suitable areas to support new populations and spread risk would be beneficial.
87597	<i>Tachyglossus aculeatus multiaculeatus</i> Kangaroo Island Echidna	Specialised diet	Yes	Yes	No	Maybe; to relocate from areas if ant mounds have been destroyed	No	No	Yes; but less critical than for other mammal species	

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267	<i>Burramys parvus</i> Mountain Pygmy-possum	Specialised diet and habitat; very susceptible to introduced predators; social	Yes	Yes	Ex situ animals already exist	No	Yes; supplementary feeding already in place in NSW, and continuing until May	Yes	Yes	
88218	<i>Antechinus argentus</i> Silver-headed Antechinus	Habitat specialist, prefers long-unburnt vegetation	Yes; resurvey previous locations using dogs	Yes	No	No	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes; cattle and horses especially in and adjacent to Kroombit Tops and Blackdown Tablelands National Parks	Yes	- Lantana and other weed control, especially post-fire
87617	<i>Mastacomys fuscus mordicus</i> Broad-toothed Rat	Specialised herbivore; very susceptible to impacts from introduced herbivores and predators; Needs long-unburnt habitat	Yes	Yes	Maybe	Unlikely; nearby unburnt habitat may be saturated or unsuitable, but translocations to recovered habitat may be possible later	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	
88	<i>Pseudomys fumeus</i> Smoky Mouse, Konoom	Very susceptible to impacts from introduced herbivores and predators	Yes	Yes	Maybe; existing ex-situ population at Priam Breeding Facility in NSW	Maybe; has disappeared from some habitat patches where it used to occur	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	- Habitat very susceptible to Phytophthora dieback, any salvage efforts and surveys should ensure no spread. - Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range
85104	<i>Phascolarctos cinereus</i> Koala	Specialised herbivore	Yes	Yes	No; ex situ not required, but in some circumstances koalas may be salvaged into care for welfare reasons	No	Yes but these actions less relevant 2-4 months post fires	No	Yes; control wild dogs in areas where canopy is not continuous; control peri-urban domestic dogs	- Avoid clearing of koala habitat
89289	<i>Notomacropus parma</i> Parma Wallaby	Grazer, habitat specialist needing mix of grass and unburnt forest Vulnerable to predation after fire clears vegetation	Yes	Yes	Maybe	Unlikely; nearby unburnt habitat may be saturated or unsuitable, but translocations to recovered habitat may be possible later	No	Yes; remove grazing introduced species	Yes; especially foxes	- Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range

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255	<i>Petaurus australis</i> Yellow-bellied Glider	Hollow-dependent, social	Yes	Yes	Maybe	Unlikely; nearby unburnt habitat may be saturated or unsuitable, but translocations to recovered habitat may be possible later	Yes; hollows (trials required to identify best approach; could help maintain viability of small populations)	No	No	- Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range
254	<i>Petauroides volans</i> Greater Glider	Specialised herbivore requiring long-unburnt habitat, and large tree hollows	Yes; northern subspecies less of a concern, although it was heavily impacted in 2018 by fire	Yes	Maybe	Unlikely; nearby unburnt habitat may be saturated or unsuitable, but translocations to recovered habitat may be possible later. Over the longer term, trials could help establish reasons for decline	Yes; hollows (trials required to identify best approach; could help maintain viability of small populations)	No	No	- Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range
225	<i>Petrogale penicillata</i> Brush-tailed Rock-wallaby	Large herbivore; very susceptible to impacts from introduced herbivores and predators	Yes; In Qld surveys are needed to identify the core populations, and their status	Yes	Unlikely; ex situ already exists	Unlikely	Yes but now unlikely; short term feeding an option until veg regenerates, but only required where fire has been intense, and unlikely to be required now	Yes	Yes; especially foxes	
66645	<i>Potorous tridactylus tridactylus</i> Long-nosed Potoroo	Very high susceptibility to introduced predators (compounded by cover loss)	Yes; there is little information on population status in Qld	Yes	Maybe; if required to create founders for later reintroductions	Maybe; previous parts of range now unoccupied. In longer term, translocations to burnt parts of range that have recovered may be possible.	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	- Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range

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75184	<i>Dasyurus maculatus maculatus</i> (SE mainland) Spotted Quoll	Very large home range	Yes	Yes	Unlikely	Maybe	Unlikely	Yes	Yes; especially foxes	- Education campaign to stop people killing quolls that are wandering for food. Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range.
96	<i>Pseudomys novaehollandiae</i> New Holland Mouse	Herbivore; very high susceptibility to introduced predators (compounded by cover loss)	Yes; there is little recent data from parts of range, including Qld	Yes	Maybe; founders may be required for later reintroductions Ex situ populations have existed recently so action known to be feasible	Maybe; previous parts of range now unoccupied. In longer term, translocations to burnt parts of range that have recovered may be possible.	Yes; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	
186	<i>Pteropus poliocephalus</i> Grey-headed Flying-fox	Frugivore	Yes	Yes	No; ex situ not required, but salvage into care for welfare reasons	No; they are sufficiently mobile	No; for wild animals	No	No	- Engage with community as flying foxes can move into adjacent urban areas following drought and fire. - Avoid clearing of important feeding or roosting habitat.
141	<i>Phoniscus papuensis</i> Golden-tipped Bat	Habitat specialist needing long-unburnt vegetation	Yes	Yes	Unlikely	No; they are sufficiently mobile	No	No	No	- Avoid clearing, including for forestry, of riparian habitat or rainforest/wet sclerophyll ecotones; avoid salvage logging in burnt parts of their range.
341	<i>Ornithorhynchus anatinus</i> Platypus	Population drought-affected leading up to fire, habitat specialist	Yes	Yes	Maybe	Unlikely	No	Yes	Yes	- Avoid clearing, including for forestry, of riparian habitat; avoid salvage logging in burnt parts of their range. - Longer term: Improve water quality, connectivity and availability of refuge pools, especially during drought. Improve riparian habitat condition with planting, weeding, erosion control.
90712	Mainland Dusky Antechinus <i>Antechinus mimites (swainsonii)</i>	Very high susceptibility to introduced predators (compounded by cover loss)	Yes	Yes	Unlikely	Maybe, especially in longer term	Maybe; artificial shelters in burnt areas could be an option in some circumstances	Yes	Yes	- Avoid clearing that results in population fragmentation for the species; forestry prescriptions adapted to prevent habitat patches becoming too small to support viable populations, and avoid salvage logging in burnt parts of its range

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Birds										
82338	<i>Anthochaera phrygia</i> Regent Honeyeater		Yes	Yes	Maybe; if starvation is evident Ex situ captive breeding facilities already exist at Taronga Zoo, Melbourne Zoo	No; they are sufficiently mobile	Unlikely	Unlikely	Unlikely	- Control for native competitors such as Noisy Miner in key locations. - Longer term: Plant food trees at massive scale. Avoid further clearing of habitat.
655	<i>Atrichornis rufescens</i> Rufous Scrub-bird	Habitat specialist (rainforest-eucalypt ecotone)	Yes; especially in the northern part of range - priority action is survey to describe extent and nature of impact	Yes	Maybe	Maybe; if suitable unoccupied sites are available	Unlikely; provision of fallen timber/habitat structure possible but untried and needs testing Likely to be less important than allowing natural recovery of leaf litter fauna and shrub growth	Unlikely	Maybe; especially cats	
64436	<i>Calyptorhynchus lathamii halmaturinus</i> Kangaroo Island Glossy Black-Cockatoo	Food specialist (Allocasuarina verticillata seeds) hollow-dependent for nesting	Yes	Yes	No; better to manage small population in situ	No; they are sufficiently mobile	Yes; erect extra nest boxes at known nest sites where breeding pairs are present	Yes; protect regenerating Allocasuarinas from grazing	Yes; protect nests from native and introduced competitors (galahs, corellas, honeybees)	- Protect revegetation of drooping she oak feeding habitat. - Avoid further clearing of habitat.
533	<i>Dasyornis brachypterus</i> Eastern Bristlebird	High predicted mortality from fire; required habitat has been declining (can thus be considered specialised)	Yes	Yes	Maybe; ex situ captive breeding facilities already exist, but there is no capacity for new founders. Northern population, especially in Qld, at very high risk of extirpation, and salvage may be the best option.	Northern population: No; potential habitat in the north has been burnt. Southern population: Yes; emergency salvage undertaken in Vic. Further analysis need to consider if, longer term, this species would benefit from translocation to other habitat.	No	Yes	Yes	- Restoration of habitat (including with careful burning) in north part of range.

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652	<i>Menura alberti</i> Albert's Lyrebird	High predicted mortality from fire	Yes	Yes	Maybe	No; nearby unburnt habitat probably saturated (but translocations to recovered habitat may be possible later)	No	Unlikely; introduced herbivores uncommon in the bird's range	Maybe; especially cats, although foxes are also present in low densities	
653	<i>Menura novaehollandiae</i> Superb Lyrebird	High predicted mortality from fire	Yes	Yes	No	No; nearby unburnt habitat probably saturated (but translocations to recovered habitat may be possible later)	No	Unlikely	Maybe	
609	<i>Monarcha melanopsis</i> Black-faced Monarch	High predicted mortality from fire	Yes	Yes	No	No; they are sufficiently mobile	No	No	No	
524	<i>Origma solitaria</i> Rockwarbler	High fire overlap, although habitat may offer some protection	Yes	Yes	No	No; unburnt habitat likely to be occupied	No	Unlikely	Maybe	
84650	<i>Pezoporus wallicus flaviventris</i> Western Ground Parrot	Very susceptible to impacts from introduced predators after fire; needs long-unburnt habitat for roosting and nesting	Yes; search potential nearby habitat refuges for any additional birds that may have escaped fires, and extend protection measures to these areas	Yes; establish additional fire management measures for remaining unburnt habitat	No	Yes; wild to wild translocation part of longer term recovery plan	No	Yes	Yes	- Root root management to prevent spread.
734	<i>Pezoporus wallicus wallicus</i> Mainland Ground Parrot	High predicted mortality from fire	Yes	Yes	Yes	Maybe; especially empty habitat to the east of known range, and later translocations may be possible to burnt habitat patches where the species has been extirpated, and cant recolonise without assistance.	No	Yes	Yes	

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525	<i>Pycnoptilus floccosus</i> Pilotbird	Habitat specialist needing some long burnt/never burnt areas	Yes	Yes	Maybe	Maybe; if surveys can identify suitable unoccupied habitat; and later translocations may be possible to burnt habitat patches where the species has been extirpated, and cant recolonise without assistance.	No	Unlikely	Maybe	
67121	<i>Zoothera lunulata halmaturina</i> Western Bassian Thrush	High predicted mortality from fire; Required habitat has been declining	Yes	Yes	Maybe	No; they are sufficiently mobile on KI; translocation to mainland sites may be possible after sufficient habitat restoration.	No	Unlikely	Maybe	- Restoration of habitat in mainland range through weed removal, revegetation, as well as rabbit and other herbivore control.
494	<i>Climacteris erythrops</i> Red-browed Treecreeper	Poor ability to flee fire and prefers long unburnt forests	Yes	Yes	No	No; unburnt habitat likely to be occupied. Later, once burnt habitat has recovered from fire, translocations to patches where the species has been extirpated, and cant recolonise without assistance.	No	No	No	
768	<i>Callocephalon fimbriatum</i> Gang-gang Cockatoo	Large group-living parrot that favours long unburnt forests and woodlands	Yes	Yes	No	No; they are sufficiently mobile	No	No	Yes; protect nests from predators (native and introduced)	- Avoid further clearing of habitat. - Improve forestry practices to identify nest hollows and protect them.

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67036	<i>Calyptorhynchus lathami lathami</i> South-eastern Glossy Black-Cockatoo	Large group-living parrot that favours long unburnt forests and woodlands, has a specialised diet, and needs large tree hollows for breeding	Yes	Yes	No	No; they are sufficiently mobile	Yes; erect extra nest boxes in areas known to have breeding pairs, and near existing food resources	Yes; protect regenerating casuarinas from grazing (this is for longer-term recovery)	No	- On-ground assessment should include mapping unburnt stands of Allocasuarina, assessing availability of hollows in proximity to these, and determining fire impacts on hollow availability Longer-term: - Re-establish food supply in areas where fire has resulted in long-term loss of Allocasuarina stands and natural regeneration will be insufficient to meet future needs; seeds will only become available after 10 years. Avoid further clearing of habitat. Improve forestry practices to identify nest hollows and protect them. - Re-establish water resources where erosion and sedimentation post fire has drained/filled in critical water holes (i.e. water rare in landscape). However, they will drink from very small water sources so only needed waterholes in very driest landscapes - which were largely unburnt this time round. If waterhole deepening is needed for this species, which can drink from smallest of puddles, it will be critical for many larger taxa as well.
81024	<i>Psophodes nigrogularis lashmari</i> Kangaroo Island Western Whipbird	High predicted mortality from fire; prefers long unburnt habitat; fire likely to increase predation risk substantially	Yes	Yes	Maybe	No; unburnt habitat likely to be occupied	No	Yes; protect regenerating vegetation from grazing if herbivores abundant at priority sites	Yes; this is the most important priority for the short-medium term	
85046	<i>Stipiturus malachurus halmaturinus</i> Kangaroo Island Southern Emu-wren	High predicted mortality from fire; prefers long unburnt habitat; fire likely to increase predation risk substantially	Yes	Yes	No	Yes; if surveys can identify suitable unoccupied habitat. Once habitat has recovered from fire, translocations to patches where the species has been extirpated, and cant recolonise without assistance.	No	Yes; protect regenerating vegetation from grazing if herbivores abundant at priority sites	Yes; this is the most important priority for the short-medium term	

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Reptiles										
			On-ground assessment of the impacts of fire of varying severity particularly important for reptiles because the population response to fire is often poorly known							
59199	<i>Eulamprus leuraensis</i> Blue Mountains Water Skink	High vulnerability to predation and herbivory, and species was declining prior to fires	Yes	Yes	Maybe; no existing captive breeding programs to our knowledge	No; in short term. Trial translocations to recovered habitat may be possible in longer term	No	Yes; pigs impacting swamp habitat	Yes; feral cats	- Protect remaining habitat from pollution, recreational vehicles, urban development, and groundwater extraction as a result of mining that leads to the drainage of swamps
1182	<i>Hoplocephalus bungaroides</i> Broad-headed Snake	Listed because of pre-fire status and overlap	Yes	Yes	Maybe; there are several holdings (10), although might be difficult to breed in captivity	No; unburnt habitat likely to be occupied	No	Maybe; grazing by feral goats identified as a threat	Maybe	- Review of private broad-headed snake ownership. Prevent illegal take from the wild. Avoid further habitat clearing/modification for residential development, or from collection of bush rock
86071	<i>Wollumbinia belli</i> (<i>Myuchelys belli</i>) Bell's Turtle, Western Sawshelled Turtle, Namoi River Turtle, Bell's Saw-shelled Turtle	Listed mainly because of pre-fire status and overlap Assumed to be vulnerable to predators and herbivores, particularly their nests	Yes	Yes	Maybe; no captive holdings to our knowledge	Maybe; could move animals upstream from affected areas if evidence of low water quality, mortality etc downstream	No	Maybe; cattle are an issue, so fencing might be more appropriate than lethal control	Maybe; predation on eggs by foxes, goannas, pigs; and on hatchlings by invasive fish; but predator control but may require very strong reduction in predator numbers to be effective	- Sedimentation an issue: consider sedimentation fencing or erosion mitigation. - Understand disease that causes blindness, and mitigate. Avoid further modification of water flows; remediate some existing barriers to water flow; avoid further degradation of riparian vegetation from clearing and agricultural uses, including access of stock.
59550	<i>Nangura spinosa</i> Nangur Spiny Skink	Prefers unburnt vegetation; fire probably amplifies threats from feral animals and weeds	Yes	Yes	Maybe; but they have a complex social structure At least one captive population already exists	No; better to manage threats around population	Maybe; potential trials of artificial burrows to evaluate their use.	Yes; red deer	Yes; feral cats, foxes	- New access tracks created around populations to suppress fire will allow ingress of weeds, feral animals, unauthorised visitation, so: Fencing to restrict access by collectors, photographers; and Weed control along new/renewed fire breaks. - Develop new advice and guideline documents which has assumed fire does not penetrate skink's habitat

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86072	<i>Wollumbinia georgesi</i> (<i>Myuchelys georgesi</i>) Georges' Snapping Turtle, Bellinger River Snapping Turtle, Georges Helmeted Turtle	Listed mainly because of pre-fire status and overlap Assumed to be vulnerable to predators and herbivores, particularly their nests	Yes	Yes	Maybe; captive breeding populations already exist at Taronga Zoo and Symbio Wildlife Park following virus outbreak and mass mortality in 2015 in home range. Ongoing release-to-wild program	Maybe; could move animals upstream from affected areas if evidence of low water quality, mortality etc downstream	No	Maybe; cattle are an issue, so fencing might be more appropriate than lethal control	Maybe; predation on eggs by foxes, goannas, pigs; and on hatchlings by invasive fish; but predator control may require very strong reduction in predator numbers to be effective	- Understand disease that causes lesions and mitigate. Avoid further modification of water flows; remediate some existing barriers to water flow; avoid further degradation of riparian vegetation from clearing and agricultural uses, including access of stock. Improve quality of riparian vegetation. Remove Emydura macquarii
64721	<i>Cyclodomorphus praealtus</i> Alpine She-oak Skink	Habitat specialist, likely high mortality during fire; predation post fire could be high	Yes	Yes	Maybe; there are existing holdings at Healesville	No; not as a salvage option.	No	Yes; habitat degradation through grazing and trampling by cattle, feral horses, deer and pigs	Yes; predation from rats, foxes, cats, dogs	- Weed control (<i>Hieracium aurantiacum</i>), and protect habitat from development, ski runs etc
59628	<i>Coeranoscincus reticulatus</i> Three-toed Snake-tooth Skink	Rainforest species that could be very sensitive to fire; predation post fire could be high	Yes	Yes	Maybe; no existing captive breeding programs to our knowledge	No; not as a salvage option.	No	Yes; loss of leaf litter and compaction of soils by stock identified as a threat, as well as habitat degradation by pigs	Yes; feral cats and foxes regarded as a threat	- Bimodal veg range (heathlands and wet forest), so impacts and actions might be different in each
90714	<i>Egernia roomi</i> , Kaputar rock skink	Extremely narrowly distributed and with >60% of habitat burned	Yes	Yes	Maybe; risky due to the very small population size	No; not as a salvage option.	Maybe; restoration of surface rocks	Yes; feral goats a potential issue	Maybe; little known about threats to the species	- Minimise human disturbance, including trampling and rock removal
83079	<i>Liopholis guthega</i> , Guthega skink	Narrow distribution; somewhat specialised diet but limited information available	Yes	Yes	Maybe; current holdings at Healesville	No; not as a salvage option.	No	Yes; habitat degradation through grazing and trampling by cattle, feral horses, deer and pigs	Yes; predation by fox and cat listed as threats	- Avoid further clearing of habitat for recreational infrastructure development (ski resorts)
64728	<i>Lampropholis elongata</i> , Long sunskink	Data deficient species that has had nearly 70% of its very narrow range affected by the fires	Yes	Yes	Maybe; no existing captive breeding programs to our knowledge	No; not as a salvage option.	No	Maybe; some suggestion grazing has a negative influence on the species, but very little known	Maybe; given its size, but highly uncertain	- Protection of habitat from land-use intensification
84408	<i>Pseudemoia cryodroma</i> , Alpine bog skink	Habitat specialist; likely has high mortality during fire; exposed to predation and herbivore impacts after fire	Yes	Yes	Maybe; no existing captive breeding programs to our knowledge	No; not as a salvage option.	No	Yes; habitat degradation by feral herbivores (horses) identified as a threat by IUCN	Yes; high densities of predators around ski resorts (near habitat) has been identified as a threat by IUCN	- Control weeds. Avoid further clearing of habitat for recreational infrastructure development (ski resorts)

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89460	<i>Myuchelys purvisi</i> Manning River Helmeted Turtle, Purvis' turtle	Predation and herbivore impacts after fire could be high	Yes; Citizen science techniques could help locate nests	Yes	Yes. A facility to research ex situ breeding programs for the species has been completed at the Australian Reptile Park	Maybe; could move animals upstream from affected areas if evidence of low water quality, mortality etc downstream	No	Yes; grazing and trampling by introduced herbivores, including by livestock, a potential threat	Maybe; predation by foxes, feral pigs identified as a threat, so management to prevent nest predation could be important (possible activity for citizen scientists)	- Avoid further modification of water flows; remediate some existing barriers to water flow; avoid further degradation of riparian vegetation from clearing and agricultural uses, including access of stock. Understand extent of hybridisation with <i>Emydura macquarii</i>
84785	Rainforest Cool-skink <i>Harrisoniascincus zia</i>	Rainforest species that may be particularly sensitive to fire impacts; may be vulnerable to increased predation from predators after fire	Yes; including populations west of the Main Range at Lamington, and the population near Springbrook, that have not been reported for over a decade	Yes	No	No; not as a salvage option.	No	Unlikely	Maybe; especially cats, foxes are at low density in the species range	- Collect genetic data to examine genetic divergence between Border Ranges and Lamington population
59203	Southern Water-skink <i>Eulamprus tympanum</i>	Habitat specialised; could suffer high mortality during fire; vulnerable to predator and herbivore impacts after fire	Yes	Yes	Unlikely	Unlikely as a salvage option	No	Yes; grazing and trampling by introduced herbivores, including by livestock, a potential threat	Maybe; predation by foxes, pigs, cats	- Avoid further habitat clearing/modification for agriculture
59160	Glossy Grass Skink <i>Pseudemoia rawlinsoni</i>	Habitat specialised; could suffer high mortality during fire; vulnerable to predator impacts after fire	Yes	Yes	Unlikely	Unlikely as a salvage option	No	Yes; grazing and trampling by introduced herbivores a potential threat	Maybe; predation by foxes, pigs, cats	- Avoid further habitat clearing/modification for agriculture or development; forestry practices modified to limit damage to habitat
89490	Moritz's Leaf-tailed Gecko <i>Saltuarius moritzi</i>	Habitat specialist that prefers long unburnt habitat	Yes	Yes	Unlikely	Unlikely as a salvage option	No	Unlikely; semi arboreal	Maybe; but often out of reach of predators	
1170	Mustard-bellied Snake <i>Drysdalia rhodogaster</i>	Fire mortality could be high, and prefers long unburnt habitat	Yes	Yes; Fire managed to retain basking areas on rocks	Unlikely	Unlikely as a salvage option	Maybe; artificial refuges could be an option, given it is known to use artificial habitats	Maybe	Maybe	

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1529	Red-tailed Calyptotis <i>Calyptotis ruficauda</i>	Somewhat specialised habitat preference; likes long unburnt habitat; potentially exposed to high predation after fire	Yes	Yes	Unlikely	Unlikely as a salvage option	Maybe; . artificial habitats could help and reduce predation pressure	Maybe	Maybe	
64743	Granite Leaf-tailed Gecko <i>Saltaurius wyberba</i>	No traits that stand out for fire vulnerability, but reasonable fire overlap for a species with a very small range	Yes	Yes	Unlikely	Unlikely as a salvage option	No	No; lives among rocks	Maybe; but often out of reach of predators	- Prevent illegal take from the wild
1683	Broad-tailed Gecko <i>Phyllurus platurus</i>	No traits that stand out for fire vulnerability, but reasonably large fire overlap for a species with a very small range	Yes	Yes	Unlikely	Unlikely as a salvage option	No	No; lives among rocks	Maybe; but often out of reach of predators	
89489	Kate's Leaftail Gecko <i>Saltuarius kateae</i>	Traits don't indicate high vulnerability to fire, but 100% of its range was burnt	Yes	Yes	Unlikely	Unlikely as a salvage option	No	Unlikely; semi arboreal	Maybe; but often out of reach of predators	
84754	Oakview leaf-tailed Gecko <i>Phyllurus kabikabi</i>	Rainforest species that may be particularly sensitive to fire impacts	Yes	Yes	No	No; not as a salvage option.	No	Yes; deer	Yes; cats and foxes	- Fencing to restrict access by collectors, photographers - Weed control along new/renewed fire breaks entering essential habitat Protection of remaining habitat from future fire impacts - Develop new advice and guideline documents which has previously assumed fire does not penetrate this habitat
1681	Ringed thin-tail Gecko <i>Phyllurus caudiannulatus</i>		Yes	Yes	No	No; not as a salvage option.	No	Maybe	Yes; cats and pigs	Weed control Develop new advice and guideline documents which has previously assumed fire does not penetrate this habitat Actions to deter illegal collectors could be considered (such as installation of fences or surveillance cameras at key locations)

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Frogs										
		Many frogs have small ranges, and had declining populations as a result of other threats, before the 2019-20 fires	On-ground assessment of the impacts of fire of varying severity particularly important for frogs, because the population response to fire is often poorly known							Actions that apply to many frog species to support longer term recovery: - Avoid further modifications to stream flow and reduction in water quality. - Avoid further clearing of habitat for agriculture, recreation or development. - Forestry prescriptions modified to avoid or minimise disturbance; including salvage logging - Prevent release of herbicides, insecticides and other chemicals into waterways. - Prevent releases of invasive predatory fish (e.g. trout, carp) into new waterways.
1973	<i>Heleioporus australiacus</i> Giant Burrowing Frog		Yes	Yes	Maybe; especially for southern population/species	No; not as a salvage option. In longer term, translocations to recovered habitat may be a possibility	No	Yes	Yes; cats and foxes	
66670	<i>Pseudophryne pengilleyi</i> Northern Corroboree Frog		Yes	Yes	Maybe; expansion of existing ex situ in terms of Evolutionarily Significant Units represented	No; not as a salvage option. In longer term, translocations to recovered habitat is a possibility	No	Yes; especially horses	Unknown	- Remove weeds (blackberry) from breeding habitat.
1935	<i>Philoria kundagungan</i> Mountain frog, Red & Yellow Mountain Frog		Yes	Yes	No; in short term. Captive breeding for longer term conservation could be scoped	No; not as a salvage option. In longer term, translocations to recovered habitat in light of climate change could be considered	Maybe	Maybe; cats, foxes, and pig control (relatively recent invader to main range)	Unknown	- Lantana control, in burnt brushbox communities until canopy closes. Possibly control invasive vines. Drought will have impacted this species severely
1827	<i>Litoria piperata</i> Peppered Tree Frog		Yes; species not recorded since 1979	Yes	Maybe; not seen since 1979, if populations are identified the need for emergency salvage could be very high	No	No	Unknown	Unknown	- Phylogenetic uncertainty between <i>Litoria piperita</i> , <i>L. pearsoniana</i> and <i>L. barringtonensis</i> need to be resolved.

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1915	<i>Pseudophryne corroboree</i> Southern Corroboree Frog		Yes	Yes	No; already implemented sufficiently Existing captive breeding programs include Taronga Zoo, Melbourne Zoo, Healesville Sanctuary and the Amphibian Research Centre	No; not as a salvage option. In longer term, translocations are part of this species conservation management	No; but supplementary watering in enclosures when needed	Yes	Unknown	
25959	<i>Litoria spenceri</i> Spotted Tree Frog		Yes	Yes	Maybe; through expansion of small-scale existing ex situ populations	Maybe; species may have low recruitment and may need translocation/captive support - analysis required	No	No	Unknown	- Remove weeds.
1889	<i>Taudactylus pleione</i> Kroombit Tinker Frog, Pleione's Torrent Frog		Yes	Yes	Maybe; longer term a trial of captive breeding would be valuable	No; not as a salvage option. In longer term, translocations to recovered habitat may be trialled	No	Yes	Unknown	- Remove weeds (blackberry) from breeding habitat
1942	Mixophyes balbus Stuttering Frog, Southern Barred Frog (in Victoria)		Yes	Yes	Maybe	No; not as a salvage option. In longer term, translocations to recovered habitat may be trialled	No	Maybe	Maybe	- Assessment of genetic diversity between and among populations
1944	Mixophyes iteratus Giant Barred Frog		Yes	Yes	No	No; not as a salvage option.	No	Maybe	Unknown	
64733	<i>Litoria littlejohni</i> Littlejohn's Tree Frog, Heath Frog		Yes; rapid assessment of impact of fire in the Blue Mountains and all populations south of Nowra into Victoria (Gippsland) is required	Yes	No	No; not as a salvage option.	No	Maybe	Unknown	- Assessment of chytrid levels at burnt verses unburnt sites Assessment of genetic diversity to identify potential bottlenecking

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25960	Mixophyes fleayi Fleay's Frog		Yes	Yes	No; we do not know about genetic variation between populations. Significant populations remain in the wild	No; not as a salvage option.	No	Yes; pigs, cattle upstream a threat as cause sedimentation to streams	Yes; cane toads may be heavily advantaged in NENSW by fire. No evidence for predation but significant competition.	- Assessment of genetic diversity between and among populations. Remove weeds from breeding habitat
78930	<i>Philoria richmondensis</i> Richmond Range Sphagnum Frog		Yes	Yes	Maybe; low numbers at known sites	No; not as a salvage option. In longer term, translocations to recovered habitat may be trialled	No	Maybe	Yes; including pigs	- Manage weeds. Other main threat is chytrid
78929	<i>Philoria pughii</i> Pugh's Frog		Yes	Yes	Unlikely; but potentially necessary pending post-fire assessment	No; not as a salvage option. In longer term, translocations to recovered habitat may be trialled	No	Maybe	Yes, pigs destroy burrows	- Good forestry management practices important
1937	<i>Philoria sphagnicola</i> Spahgnum Frog		Yes	Yes	Maybe	No; not as a salvage option.	No	Maybe	Unknown	
1807	<i>Litoria subglandulosa</i> New England Tree Frog		Yes	Yes	Maybe	No; not as a salvage option.	No	No	Unknown	
78964	Litoria daviesae Davies' Tree Frog		Yes	Yes	Maybe	No; not as a salvage option.	No	No	Unknown	

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Fish										
		Many fish species have small ranges, and were already impacted by drought before the 2019-20 fires								Actions that apply to many fish species to support longer term recovery: - Consistent conservation assessment (e.g. SEAP) across freshwater fish, including the priority species - Careful management of fire suppression activities (e.g. limit input of retardant and foams and soil disturbance from machinery) - Avoid further modifications to stream flow and reduction in water quality - Avoid further clearing of riparian vegetation for agriculture, recreation or development - Forestry prescriptions modified to avoid or minimise disturbance that increases sedimentation; including salvage logging - Avoid release of herbicides, insecticides and other chemicals into waterways - Remove existing invasive fish and prevent further releases of invasive fish (e.g. trout, carp, gambusia) into new waterways
26180	Honey Blue-eye <i>Pseudomugil mellis</i>		Yes	Yes	Yes	Yes; and longer term, translocate back to wild after streamside vegetation has recovered	No	No	Yes; for competitors like Eastern Gambusia if present, but not for predators	- Riparian weed control
90732	Blue Mountains Perch <i>Macquaria</i> sp nov 'hawkesbury taxon'		Yes; surveys required in all three catchments of the Hawkesbury Nepean including a population assessment and assessment of potential refuge holes for translocation	Yes	Yes	Maybe; there is potential to secure individuals in nearby waterways or dams such as Cordeaux Dam for release into wild once streamside vegetation has recovered	No	No	No	- Formal description of species

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26170	Eastern Freshwater Cod, Clarence River Cod <i>Maccullochella ikei</i>		Yes	Yes	Yes but now unlikely; some salvage done, but now likely too late for further salvage following widespread rainfall and ash input	Yes but now unlikely; following widespread rainfall and ash input. Translocations may be possible later when streamside vegetation as recovered	Maybe; provision of artificial spawning sites if sedimented stream habitat has not recovered by mid year	No	No	
64468	Oxleyan Pygmy Perch <i>Nannoperca oxleyana</i>		Yes	Yes	Yes but now unlikely; some salvage done, but now likely too late for further salvage following widespread rainfall and ash input	No; salvage to other waterways is not a priority for this species. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	No	- Reduction of alien Gambusia numbers (but very difficult to do) - Riparian weed control - A range of other long term recovery actions for this species in the recovery plan
90826	Macquarie Perch <i>Macquaria australasica (MDB)</i>		Yes	Yes	Yes but now unlikely; some salvage done, but now likely too late for further salvage following widespread rainfall and ash input	Unlikely; surveys of nearby habitat to determine suitability and if populations are already present is required first. Translocations from captive stock back to wild may be possible later when streamside vegetation ahs recovered.	No	No	No	- Need national breeding and stocking program
84745	Flathead Galaxias <i>Galaxias rostratus</i>		Yes	Yes	No; so difficult to locate that not feasible	No	No	No	No	- Develop eDNA probe to improve detection probability so that populations can be discovered for ongoing management

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90731	Yalmy Galaxias Galaxias sp nov 'yalmy'		Yes	Yes	Yes but now unlikely; salvage attempted but insufficient fish caught. Captive breeding, reinforcement stocking and establishing new populations after streamside vegetation has recovered is a priority	No; not as a salvage option because suitable translocation sites need to be located first. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in entire Rodger River catchment - Formal description of species
87177	McDowall's Galaxias Galaxias mcdowalli		Yes	Yes	No; sufficient salvage already done	No; none available. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in entire Rodger River catchment - Locate suitable translocation sites
87880	Short-tail Galaxias <i>Galaxias brevissimus</i>		Yes	Yes	Yes; some salvage done for 1 sub-population but still required for other sub-population	No; suitable translocation sites need to be located first . Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; but only if predators have breached predator barrier and invaded galaxiid habitat	- Avoid salmonid stocking in catchment until status of remaining populations assessed

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87174	East Gippsland Galaxias <i>Galaxias aequipinnis</i>		Yes	Yes	No; sufficient salvage already done	No; none available as a salvage option. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in entire Goolengook River catchment
87879	Stocky Galaxias <i>Galaxias tantangara</i>		Yes	Yes	No; salvage already conducted. Captive breeding, and establishing new populations, is a priority	No; suitable translocation sites need to be located first . Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	Yes; horses	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in entire Tantangara Creek catchment
87176	Dargo Galaxias <i>Galaxias mungadhan</i>		Yes	Yes	No; salvage already conducted	No; none available as a salvage option. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in entire upper Dargo River catchment upstream of Miners Flat Track
87175	Roundsnout Galaxias <i>Galaxias terenasus</i>		Yes	Yes	Yes; no salvage done to date and captive breeding may be required to re-establish populations	No; suitable translocation sites need to be located first. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in Genoa River catchment
81530	Non-parasitic Lamprey <i>Mordacia praecox</i>		Yes	Yes	No; marine migratory so not necessary/feasible	No	No	No	No	

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90722	SW Victorian Blackfish Gadopsis sp nov 'Western Victoria'		Yes	Yes	Yes	Maybe; but may need habitat restoration first	No	No	No	- Undertake translocations, after introduction of woody habitat - Formal description of species
90782	Cann Galaxias Galaxias sp 17 'Cann'		Yes	Yes	Yes; some salvage done but insufficient	No; suitable translocation sites need to be located first. Translocations from captive stock back to wild may be possible later when streamside vegetation has recovered, and if captive breeding successful.	No	No	Maybe; only if predators (Trout) have invaded upstream into galaxiid population	- Avoid salmonid stockings in Cann River catchment upstream of Noorinbee - Formal description of species
Crayfish										
		Most crayfish have small ranges; fire overlaps can therefore be very high								Actions that apply to many crayfish species to support longer term recovery: - Consistent conservation assessment (e.g. SEAP) across freshwater fish, including the priority species - Careful management of fire suppression activities (e.g. limit input of retardant and foams and soil disturbance from machinery) - Avoid further modifications to stream flow and reduction in water quality - Avoid further clearing of riparian vegetation for agriculture, recreation or development - Forestry prescriptions modified to avoid or minimise disturbance that increases sedimentation; including salvage logging - Avoid release of herbicides, insecticides and other chemicals into waterways - Remove existing invasive fish and prevent further releases of invasive fish (e.g. trout, carp, gambusia) into new waterways
83136	East Gippsland Spiny Crayfish <i>Euastacus bidawalus</i>		Yes	Yes	No; some salvage already undertaken, and wild-to-wild translocations possible in the future	No; most of range fire-affected, so salvage translocations unlikely to be feasible	No	Yes; horses, pigs	Maybe; pigs, foxes plus predatory fish (Trout & Redfin)	

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83138	Ellen Clark's Crayfish <i>Euastacus clarkae</i>		Yes	Yes	Yes	No; Small range 100% fire-affected, so salvage translocations unlikely to be feasible	No	Yes; horses, pigs	Maybe; pigs, foxes plus predatory fish (Trout & Redfin)	
83139	Clayton's Spiny Crayfish <i>Euastacus claytoni</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes plus predatory fish (Trout & Redfin)	
83140	Alpine Crayfish <i>Euastacus crassus</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses	Maybe; pigs, foxes plus predatory fish (Trout & Redfin)	
83141	Mud Gully Crayfish <i>Euastacus dalagarbe</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
66782	Orbost Spiny Crayfish <i>Euastacus diversus</i>		Yes	Yes	Yes; some already conducted (but insufficient)	No; most of range fire-affected, so salvage translocations unlikely to be feasible.	No	Yes; horses, pigs	Maybe; pigs, foxes plus predatory fish (Trout & Redfin)	
83144	Gamilaroi Spiny Crayfish <i>Euastacus gamilaroi</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
83146	Smooth Crayfish <i>Euastacus girurmulayn</i>		Yes	Yes	Yes; emergency rescue to ex-situ	No, but maybe later; has a very small range, extent of fire overlap needs to be resolved with on ground survey before feasibility of salvage translocation can be considered	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
68363	Bloodclaw Crayfish <i>Euastacus gumar</i>		Yes	Yes	Yes; some already conducted (but insufficient)	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	

SPRAT ID	Species identified as the highest priority for urgent management action ^a		Rapid on-ground assessment for species and communities of concern ^b	Careful management of unburnt areas ^c	Emergency salvage of plant and animal species for ex-situ conservation or wild-to-wild translocation ^d		Supplementary shelter, food, and water for animals ^f	Feral predator and herbivore control to reduce the pressure on native species where appropriate		Longer-term additional actions ^g
	Common name	High scores for ecological traits			Salvage/capture into ex situ ^e	Salvage/translocate to nearby unburnt habitat or havens (including waterways for fish) ^e	Provision of supplementary resources	Exclude herbivores from unburnt and regenerating vegetation	Control introduced predators	
81538	Tianjara Crayfish <i>Euastacus guwinus</i>		Yes	Yes	Yes; emergency rescue to ex-situ	No, but maybe later; has a very small range, extent of fire overlap needs to be resolved with on ground survey before feasibility of salvage translocation can be considered	No	Yes; horses, pigs	Maybe; pigs, foxes and other crayfish (e g yabbies) out of their natural range	
83148	Blue-Black Crayfish <i>Euastacus jagabar</i>		Yes	Yes	Yes; emergency rescue to ex-situ	No, but maybe later; has a very small range, extent of fire overlap needs to be resolved with on ground survey before feasibility of salvage translocation can be considered	No	Yes; horses, pigs	Maybe pigs, foxes; cane toad also listed as a potential threat	
86602	Jagara Hairy Crayfish <i>Euastacus jagara</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
90715	Morgan's Crayfish <i>Euastacus morgani</i>		Yes	Yes	Yes; emergency rescue to ex-situ	No, but maybe later; has a very small range, extent of fire overlap needs to be resolved with on ground survey before feasibility of salvage translocation can be considered	No	Unknown	Unknown; threats poorly understood	
83173	Hairy Cataract Crayfish <i>Euastacus pilosus</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
83153	Many-bristled Crayfish <i>Euastacus polysetosus</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
83155	Riek's Spiny Crayfish <i>Euastacus riei</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
83156	Small Mountain Crayfish <i>Euastacus simplex</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	

SPRAT ID	Species identified as the highest priority for urgent management action ^a		Rapid on-ground assessment for species and communities of concern ^b	Careful management of unburnt areas ^c	Emergency salvage of plant and animal species for ex-situ conservation or wild-to-wild translocation ^d		Supplementary shelter, food, and water for animals ^f	Feral predator and herbivore control to reduce the pressure on native species where appropriate		Longer-term additional actions ^g
	Common name	High scores for ecological traits			Salvage/capture into ex situ ^e	Salvage/translocate to nearby unburnt habitat or havens (including waterways for fish) ^e	Provision of supplementary resources	Exclude herbivores from unburnt and regenerating vegetation	Control introduced predators	
90716	Arte Spiny Crayfish <i>Euastacus</i> sp 1		Yes	Yes	Yes	No; all of range impacted so no options now, but there may be options later	No	Unknown	Yes; predatory fish (trout)	- Formal description of species
90717	Cann Spiny Crayfish <i>Euastacus</i> sp 2		Yes	Yes	Yes	No; all of range impacted so no options now, but there may be options later	No	Unknown	Yes; predatory fish (trout)	- Formal description of species
83168	West Snowy Spiny Crayfish <i>Euastacus</i> sp 3		Yes	Yes	Yes	No; all of range impacted so no options now, but there may be options later	No	Unknown	Yes; predatory fish (trout)	- Formal description of species
83157	Small Crayfish <i>Euastacus spinichelatus</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	
83158	Sutton's Crayfish <i>Euastacus suttoni</i>		Yes	Yes	Maybe; need to assess feasibility	Maybe; wild-to-wild translocations possible in the future	No	Yes; horses, pigs	Maybe; pigs, foxes; cane toad also listed as a potential threat	- Possibly look at illegal harvest